Cold chain sabotage Iranian IRGC disrupted refrigeration controllers at 700 French retail locations, forcing emergency restocks and risking food spoilage. How many of your OT assets are truly air-gapped?

🔒 VAPT TAKEAWAY The attack leveraged a classic PLC firmware tampering vector: attackers gained a remote shell on the controller. They altered the defrost cycle and left a backdoor for later RCE. The breach was not a phishing story; it was a supply-chain intrusion. It bypassed perimeter firewalls because the devices sit on the same LAN as POS terminals. In our pen-tests, we see the same pattern: “air-gap” is a myth when SCADA networks share VLANs with business IT.

$ **COLD CHAIN SABOTAGE**

⚠ TECHNICAL RADIANCE What actually happens: “If any workstation on the same VLAN is compromised, the attacker can pivot to PLCs using default credentials.” They can also exploit known CVE-2025-4620 for industrial Ethernet stacks. In the French case, the adversary entered via a compromised vendor VPN account, then used the Modbus default admin password to inject malicious code. The result? A silent RCE that only manifested when the defrost schedule hit, giving the attackers a 48-hour window

🔒 KEY ARCHITECTURE - Enforce Zero-Trust segmentation between IT and OT, not just VLAN separation. - Deploy SIEM correlating temperature sensor anomalies with login events. - Integrate SOAR playbooks that auto-isolate a controller when its firmware hash deviates from baseline. Patch it now.

🔒 PRACTITIONER MICRO-STORY Last week our team ran a red-team exercise on a grocery chain’s cold-storage network. I watched the console flash as a compromised admin account silently re-programmed a PLC to raise the freezer temperature by 5 °C. Within minutes the alert queue filled with “temperature out-of-range” events that our SIEM had missed because the log source was mis-tagged as “environmental”. We learned fast.

[NO GLYPH] OPERATIONAL CHECKLIST • Verify every PLC firmware version against vendor hash list. • Rotate default credentials on all industrial devices.

If you still trust VLANs to protect your cold chain, expect the next breach to hit your shelves.